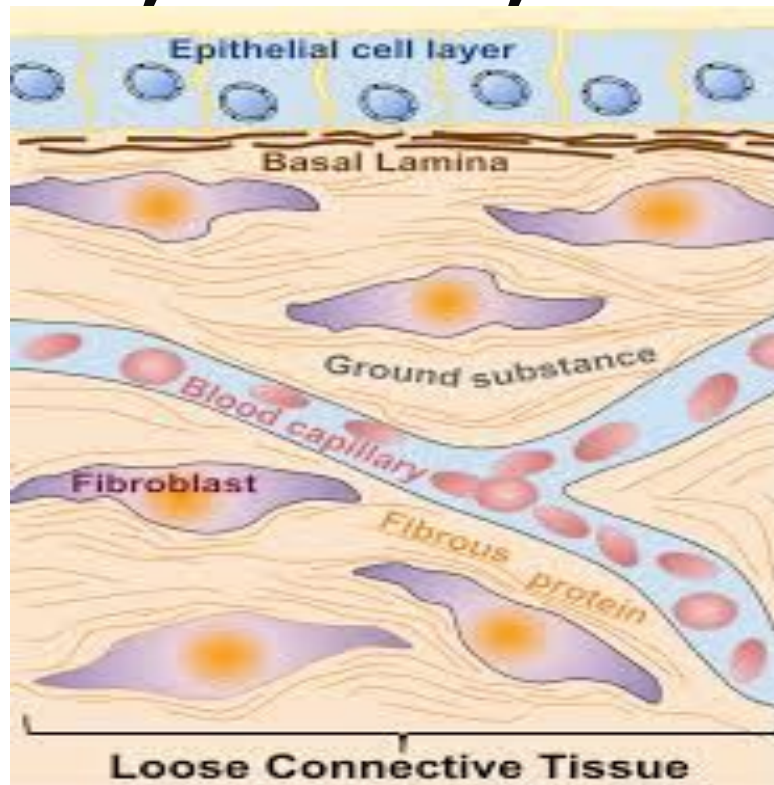


THE EXTRACELLULAR MATRIX & SPECIALIZED CONNECTIVE TISSUES OF BONES, GUMS, AND TEETH.



Dr. Humaira Aman Ali

LEARNING OBJECTIVES

- Understand the structure and functions of the extracellular matrix (ECM).
- Explain major components of ECM.
- Describe heteropolysaccharides and glycosaminoglycans (GAGs).
- Relate ECM biochemistry to dental tissues and clinical relevance.

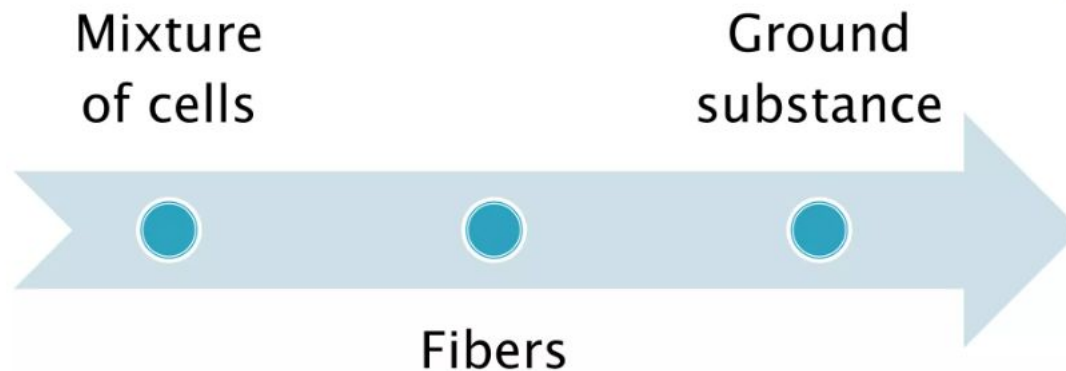
CONNECTIVE TISSUE

The connective tissue is formed by:

CELLS

and

EXTRACELLULAR MATRIX
(intercellular matrix)



CELLS OF CONNECTIVE TISSUE

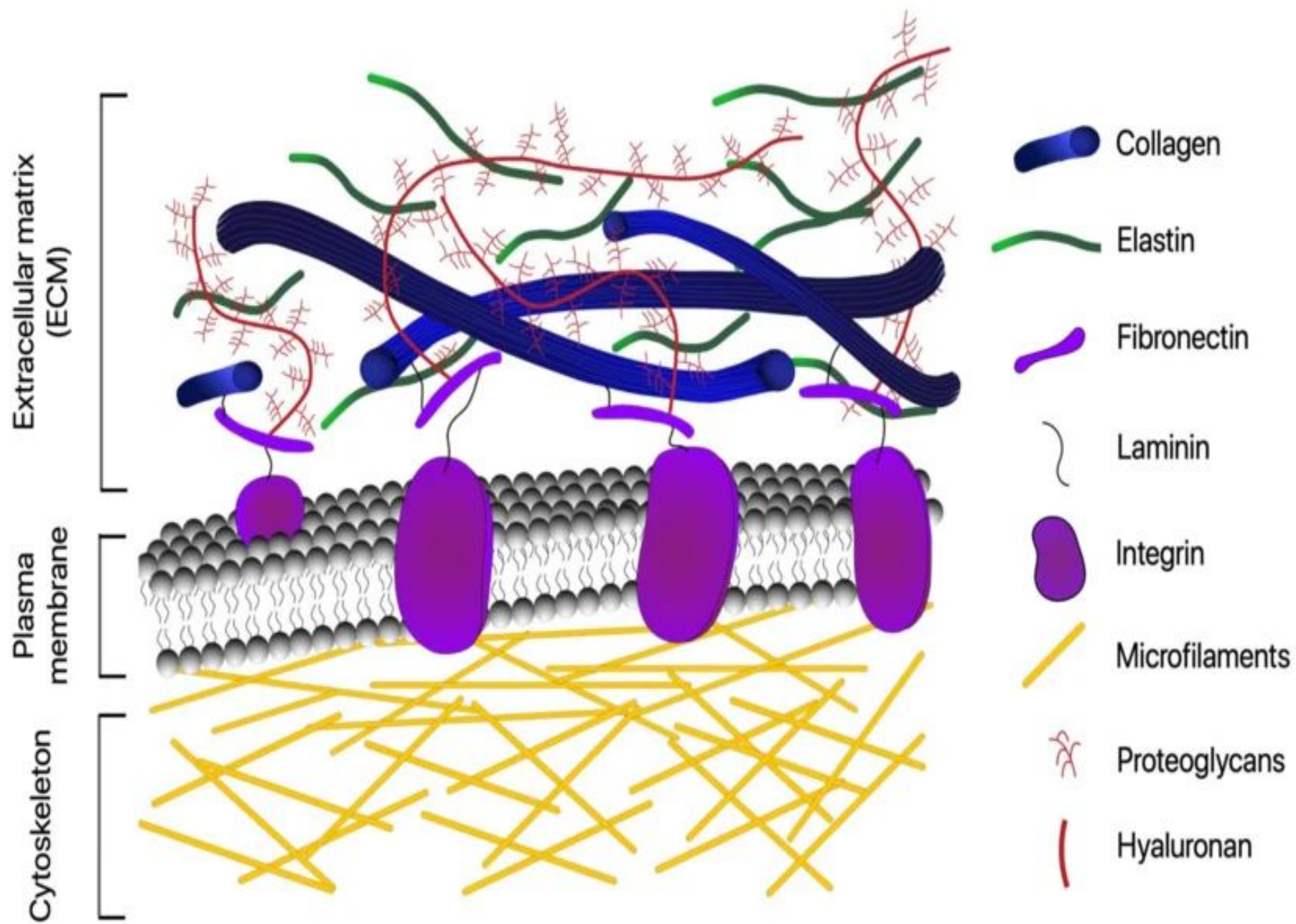
- Fibroblasts
- Chondroblasts (cartilage)
- Osteoblasts (bone)
- Odontoblasts (tooth)

These cells synthesise extracellular matrix.

WHAT IS THE EXTRACELLULAR MATRIX?

Definition:

- “The extracellular matrix is **a complex network of proteins, glycoproteins, and polysaccharides** that provide structural and biochemical support to surrounding tissues”.



EXTRACELLULAR MATRIX

- Complex network of macromolecules **outside cells**.
- Provides **structural support** and **biochemical signaling**.
- Essential for **tissue integrity** and **cell communication**.
- Highly important **in connective tissues and dental structures**.

PARTS OF THE EXTRACELLULAR MATRIX

- FIBRILLAR PROTEINS (collagen, elastin)
 - insoluble in water
- GLYCOPROTEINS (e.g. fibronectin, laminin)
- GLYCOSAMINOGLYCANS AND PROTEOGLYCANS (ground substance)
 - soluble in water, easily hydrated



Saccharide content
increases

FUNCTIONS OF MAJOR COMPONENTS OF ECM

Components of Interstitial matrix- ECM

FIBROUS STRUCTURAL PROTEINS

Confers tensile strength and recoil.

- Collagen
- Elastin

WATER HYDRATED GELS

Permits compressive resistance and lubrication

- Proteoglycans
- Hyaluronans

ADHESIVE GLYCOPROTEINS

Connects ECM elements to one another and to cells.

- Fibronectin
- Laminin

FIBRILLARY PROTEINS

- Structural proteins

- Collagen *firmness*
 elasticity
- Elastin

COLLAGEN

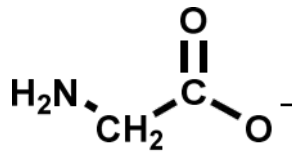
- Most abundant protein in the human body. They form approximately 25 % of all body proteins.
- Main protein of the extracellular matrix.
- Component of tendons, cartilages, bones, and teeth (dentin and cement), skin and vessels.
- Provides tensile strength.
- From the nutritional point of view not fully valuable or an incomplete protein, b/c it doesn't contain tryptophan.

PRIMARY STRUCTURE OF COLLAGEN

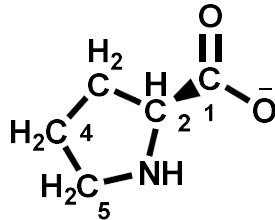
- Triple helix structure rich in glycine, proline, hydroxyproline.

Fundamental AA

▣ Glycine

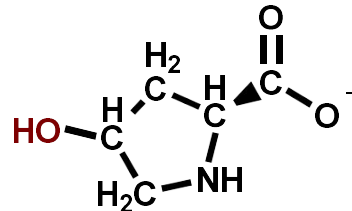


▣ Proline

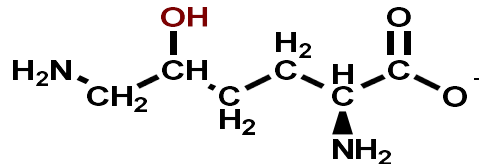


• Derived AA

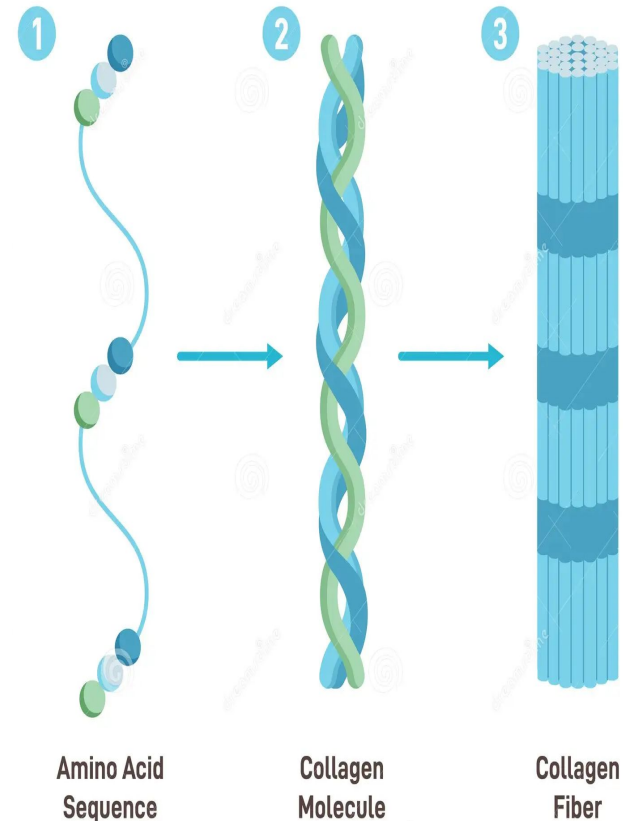
▣ 4-Hydroxyproline



▣ 5-Hydroxylysine



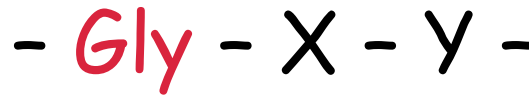
Collagen Structure



CONT..

Characteristic AA sequence

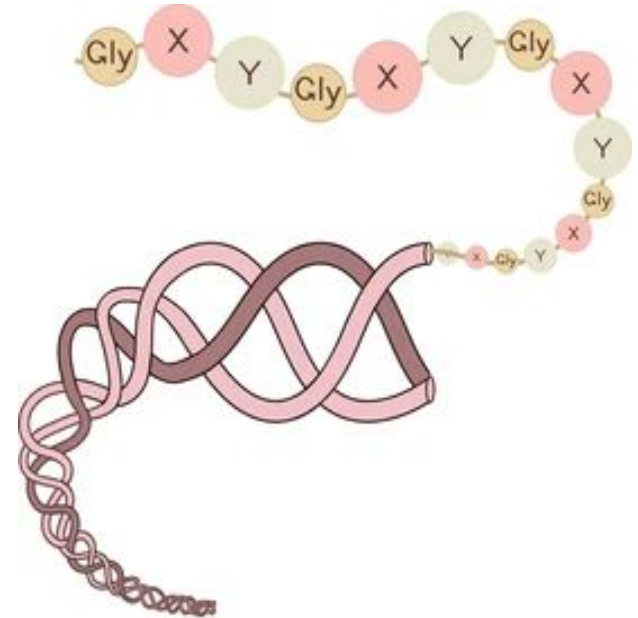
Triplet




Glycine


Proline

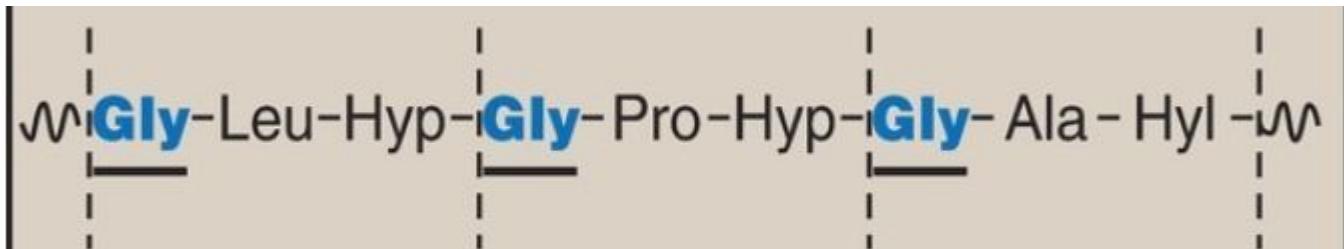

Hydroxyproline
(Hydroxylysine)



shutterstock.com · 2657511745

- A repetitive Gly-X-Y pattern, where Glycine (Gly) appears at **every third position**.
- Proline (Pro) often occupies the X position, and Hydroxyproline (Hyp) frequently occupies the Y position.

This structural motif (Gly-Pro-X or Gly-X-Hyp) enables the formation of a tight triple helix.



SECONDARY STRUCTURE OF COLLAGEN

Comparison of collagen helix to the α -helix.

Collagen helix

- **levorotatory** helix
- 3 AA/turn
- intrachain hydrogen bonds not present

α -helix

(the most common secondary structure in proteins)

- **dextrorotatory** helix
- 3.6 AA/turn
- stabilization by intrachain hydrogen bonds



Alpha helix

Alpha-helix	Collagen triple helix
Single chain	Three chains
Globular proteins & fibrous proteins (e.g. keratin)	Collagen
H-bonds <i>within</i> chain	H-bonds <i>between</i> chains
3.6 residues/turn	3 residues/turn
Right-handed helix	Left-handed helix
	Right-handed superhelix
Almost any sequence (NOT proline)	- Gly - X - Y - Gly - X - Y - X= mainly proline Y= mainly hydroxy-proline

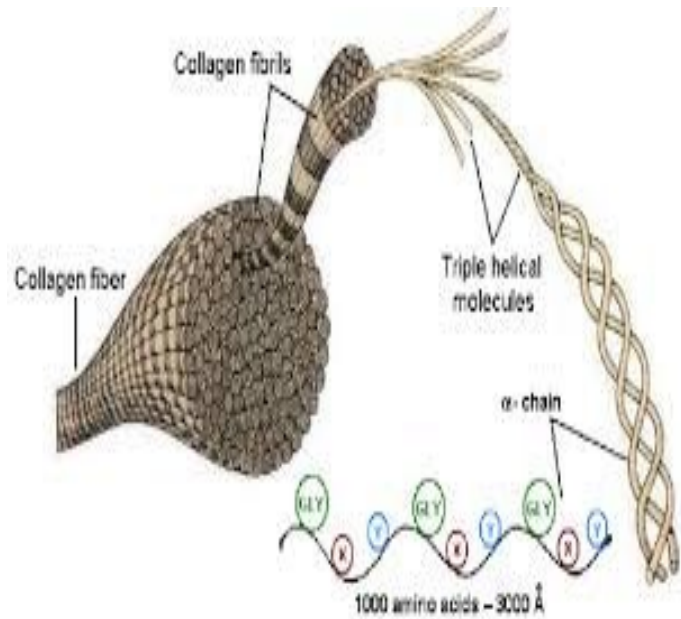


Collagen helix

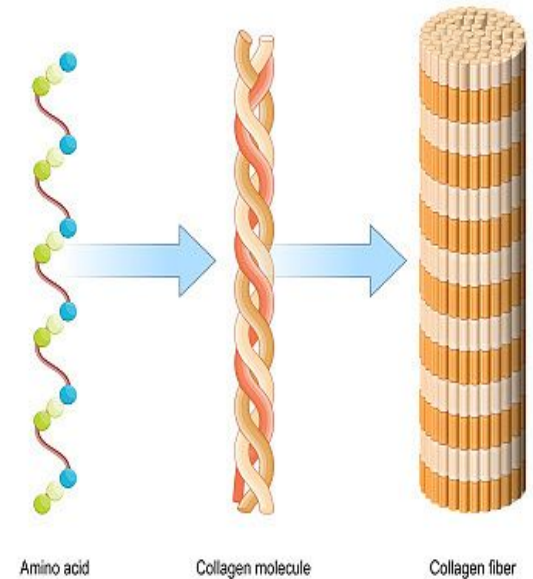
TRIPLE HELIX

Three α -chains of
collagen

gettyimages® Collagen



Triple helix
Relatively rigid

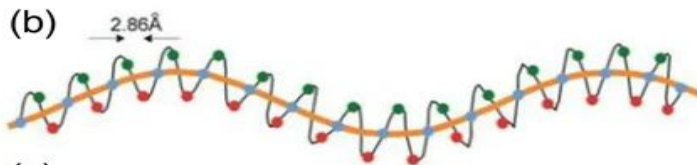
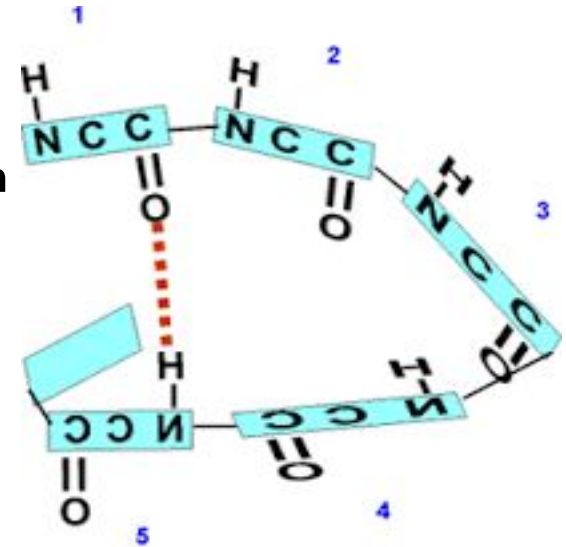


This structure is responsible for the tensile strength.

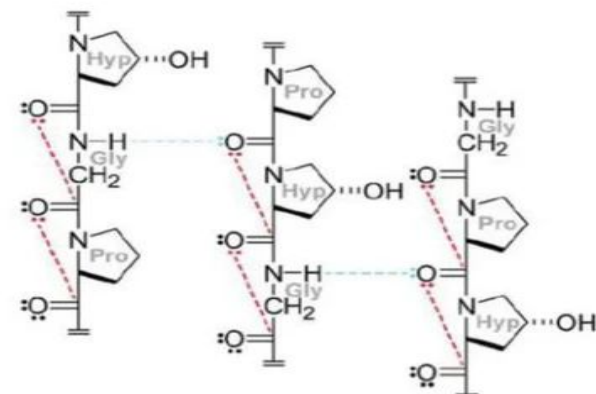
HYDROGEN BONDS IN ALPHA HELIX VS TRIPLE HELIX

Intramolecular hydrogen bonds that form between the carbonyl oxygen of one amino acid and the amide hydrogen of another, typically four residues ahead ($i + 4$ pattern).

Triple helix is stabilized by hydrogen bonds between peptide bond -NH group of glycine and C=O group of the peptide bond of the adjacent polypeptide chain.



(d)



COLLAGEN CHAINS

The collagen chain is extraordinarily long.

The collagen chains are called $\alpha 1 - \alpha 3$.

- They differ in AA representation
- Products of different genes – e.g. $\alpha_{1(I)}$ or $\alpha_{2(V)}$
- Roman digit labels the collagen type

More than 30 different types of collagen exists.

COLLAGEN CHAINS

The representation of chains differs in individual types of collagens.

The collagens may form homotrimers or heterotrimers.

Homotrimers

- molecule of collagen is formed by three identical chains;
- e.g. collagen type III is formed by three $\alpha_{1(\text{III})}$ chains

Heterotrimers

- molecule of collagen is formed by different chains;
- e.g. collagen type I is assembled of two $\alpha_1(\text{I})$ chains and one $\alpha_2(\text{I})$ chain



COLLAGEN SYNTHESIS

Collagen is an example of a protein, whose synthesis is connected with many **posttranslational modifications** (treatment of the polypeptide chain), which take part **intra- and extracellularly.**

SYNTHESIS AND POSTTRANSLATIONAL MODIFICATIONS OF COLLAGEN

Intracellular processes

Synthesis of polypeptide chain
Hydroxylation of proline and some lysine residues

Glycosylation of selected hydroxylysine residues

Formation of -S-S- bonds in extension peptides

Triple helix formation

Secretion of procollagen

POSTTRANSLATIONAL MODIFICATIONS OF COLLAGEN

Extracellular processes



↓

Proteolytic removal of propeptides
Assembling of collagen fibrils

↓

Formation of cross-links

↓

Posttranslational modifications in the course of collagen synthesis

INTRACELLULAR
PROCESSES

HYDROXYLATION OF PROLINE AND LYSINE RESIDUES

Enzymatically catalyzed reaction

- Prolylhydroxylase
- Lysylhydroxylase

Dioxygenases
contain Fe

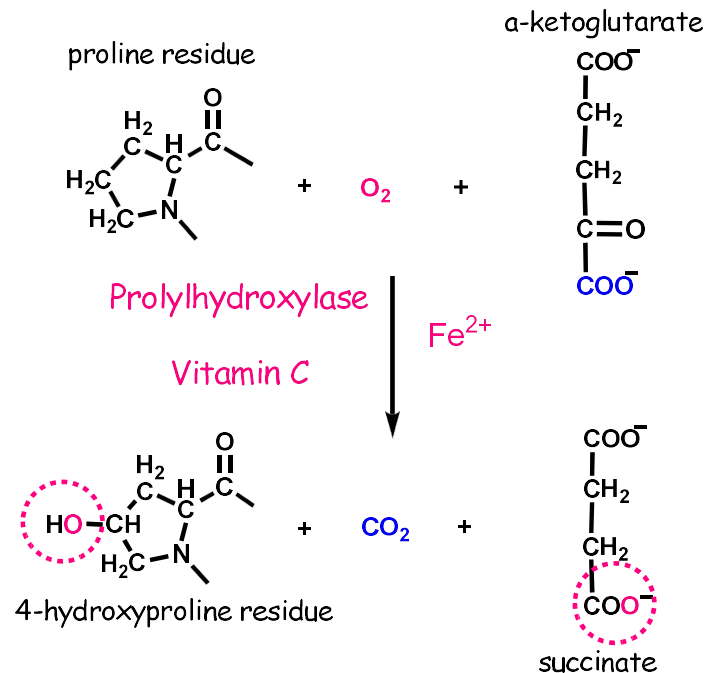
cofactors

- Vitamin C !!!
- α -ketoglutarate

Reaction needs oxygen. One O atom forms -OH group of hydroxyproline, the other becomes part of the originating succinate.

HYDROXYLATION OF THE PROLINE AND LYSINE RESIDUES

Reactions catalyzed by prolylhydroxylase



Dioxygenase
contains Fe

Vitamin C
Maintains Fe²⁺ in a
reduced state

Reaction catalyzed by prolylhydroxylase

-
- N-end -Gly-Pro-Ser-Gly-Pro-Pro-Gly-Leu- C-end
- ↓ × ×
- ↓ ↓ ↓ (Hydroxylation)
- Gly-Pro-Ser-Gly-Pro-Hyp-Gly-Leu-

HYDROXYLATION OF PROLINE AND LYSINE

Importance of proline and lysine residues hydroxylation

Hydroxyproline

- necessary for origin of triple helix by formation of hydrogen bonds between individual chains

Hydroxylysine

- glycosylation on the formed -OH group

DEFICIENCY OF VITAMIN C

Nonhydroxylated chain is not able to mature
The stable triple helix cannot be formed

Immediate degradation inside the cell

Loss of collagen in the matrix

Falling out of teeth
Vascular fragility
Poor wound healing

VITAMIN C DEFICIENCY

Avitaminosis - scurvy

Manifestation of avitaminosis in oral cavity

- swollen reddish gums
- falling out of the teeth

GLYCOSYL ATION

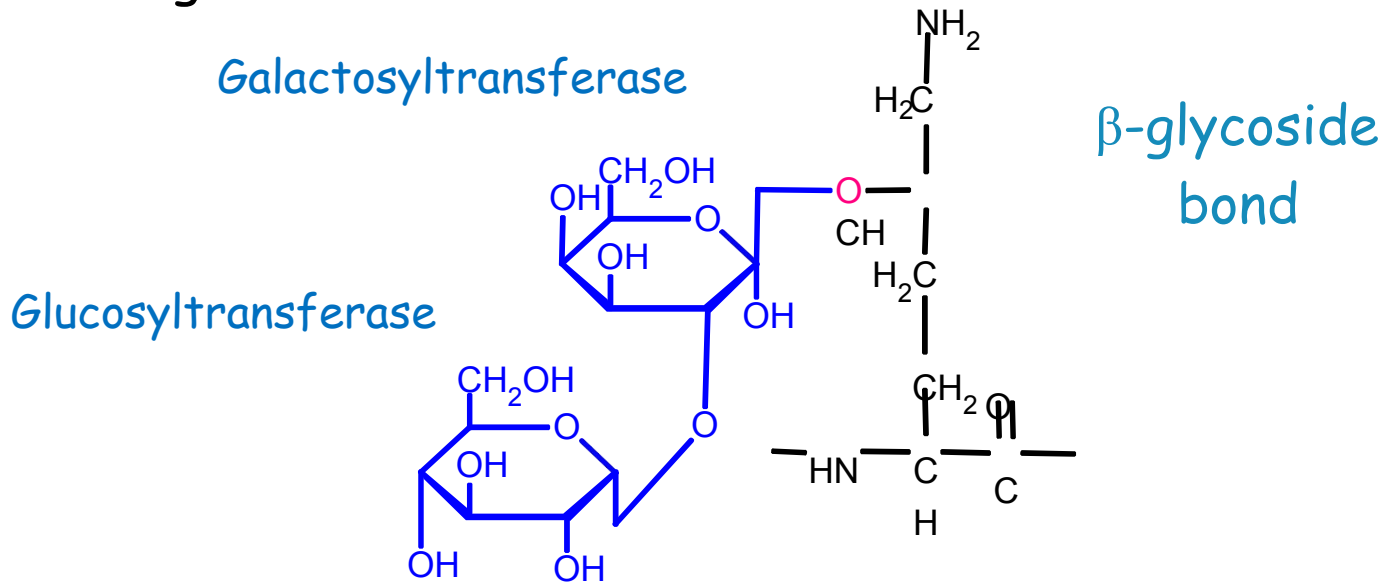
- Attachment of galactose or galactosylglucose to **-OH group of the hydroxylysine**

Enzymatically catalyzed reaction

- Galactosyltransferase
- Glucosyltransferase

GLYCOSYL ATION

Glycosylated residue of hydroxylysine in the molecule of collagen



FORMATION OF –S-S- BONDS

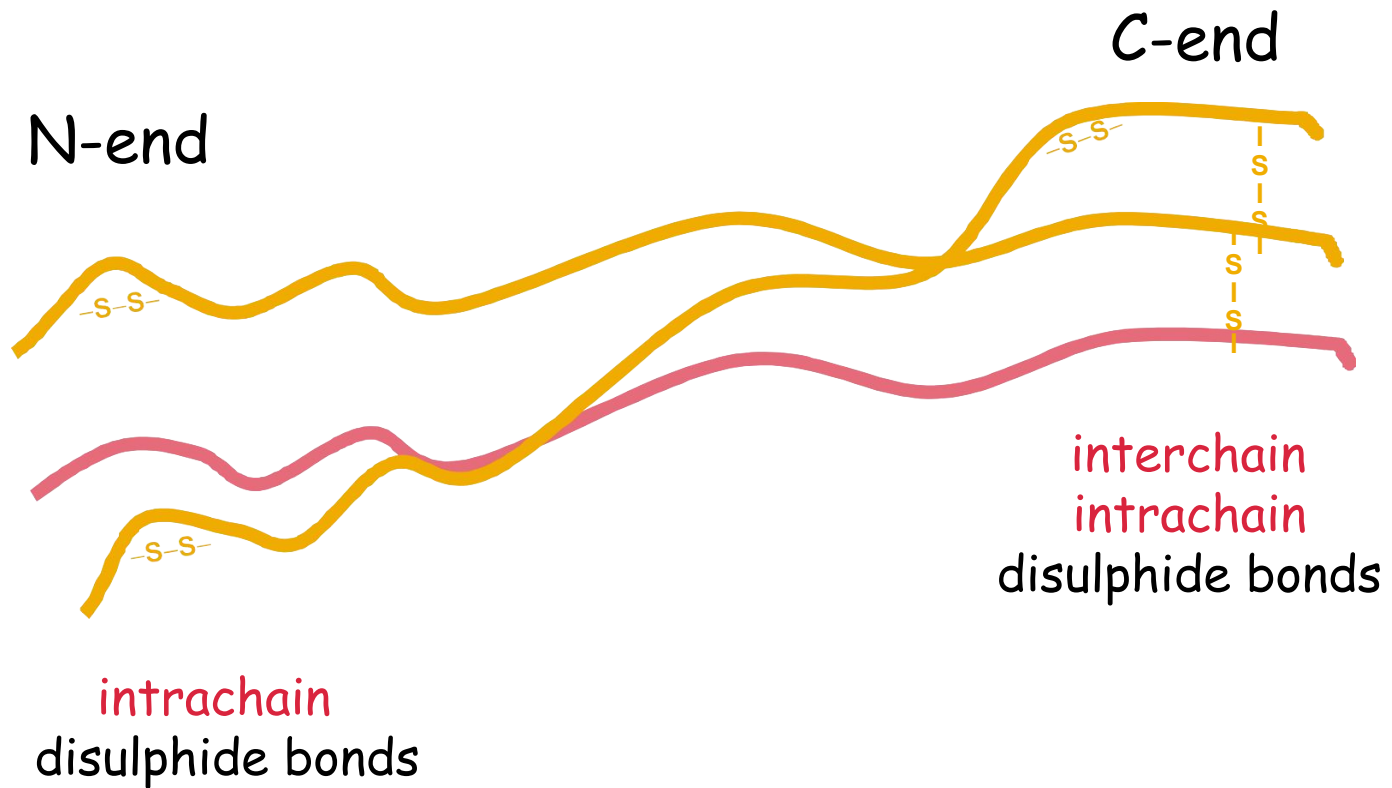
Disulphide bonds

- in the region of C-terminal propeptides
 - **interchain and intrachain** disulphide bonds
- in the region of N-terminal propeptides
 - **intrachain** disulphide bonds

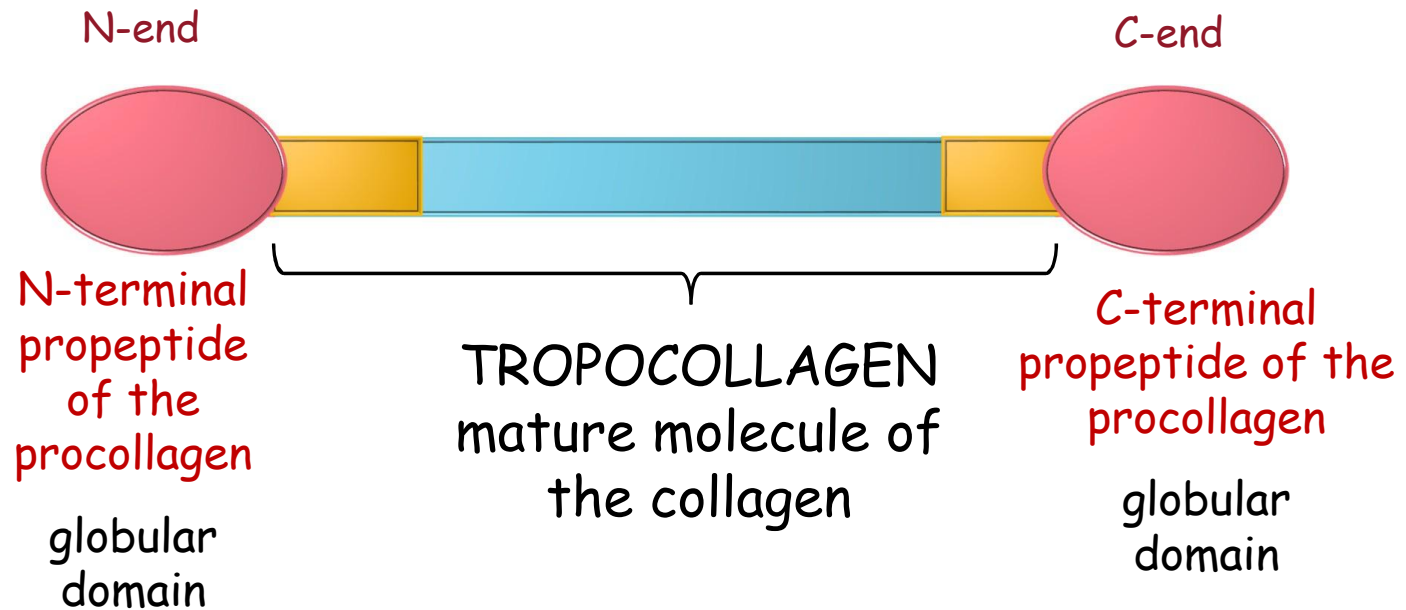
Importance

- necessary for initiation of triple helix formation
 - starts from the C-end
- secretion out of the cell

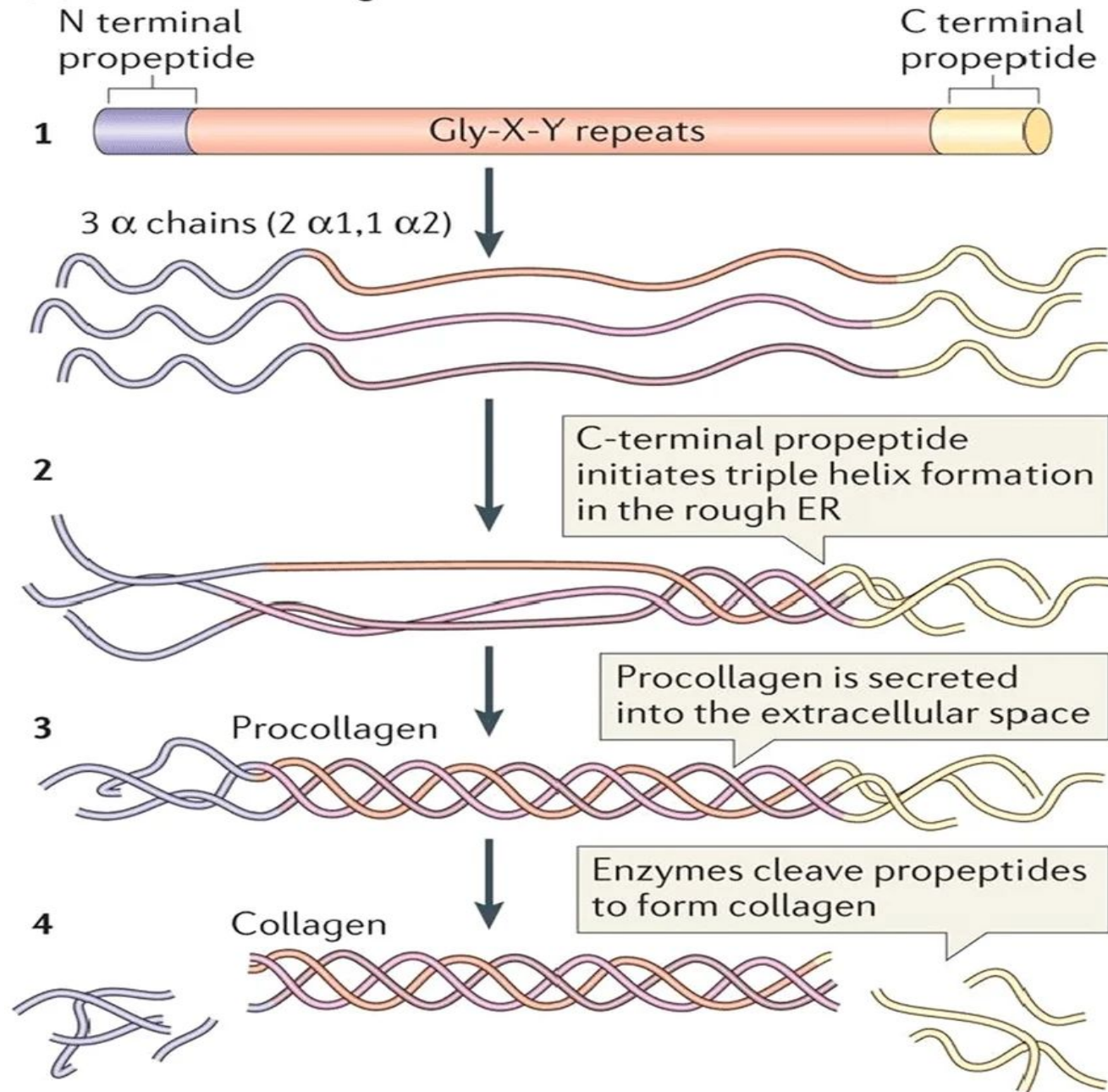
FORMATION OF -S-S- BONDS



PROCOLLAGEN



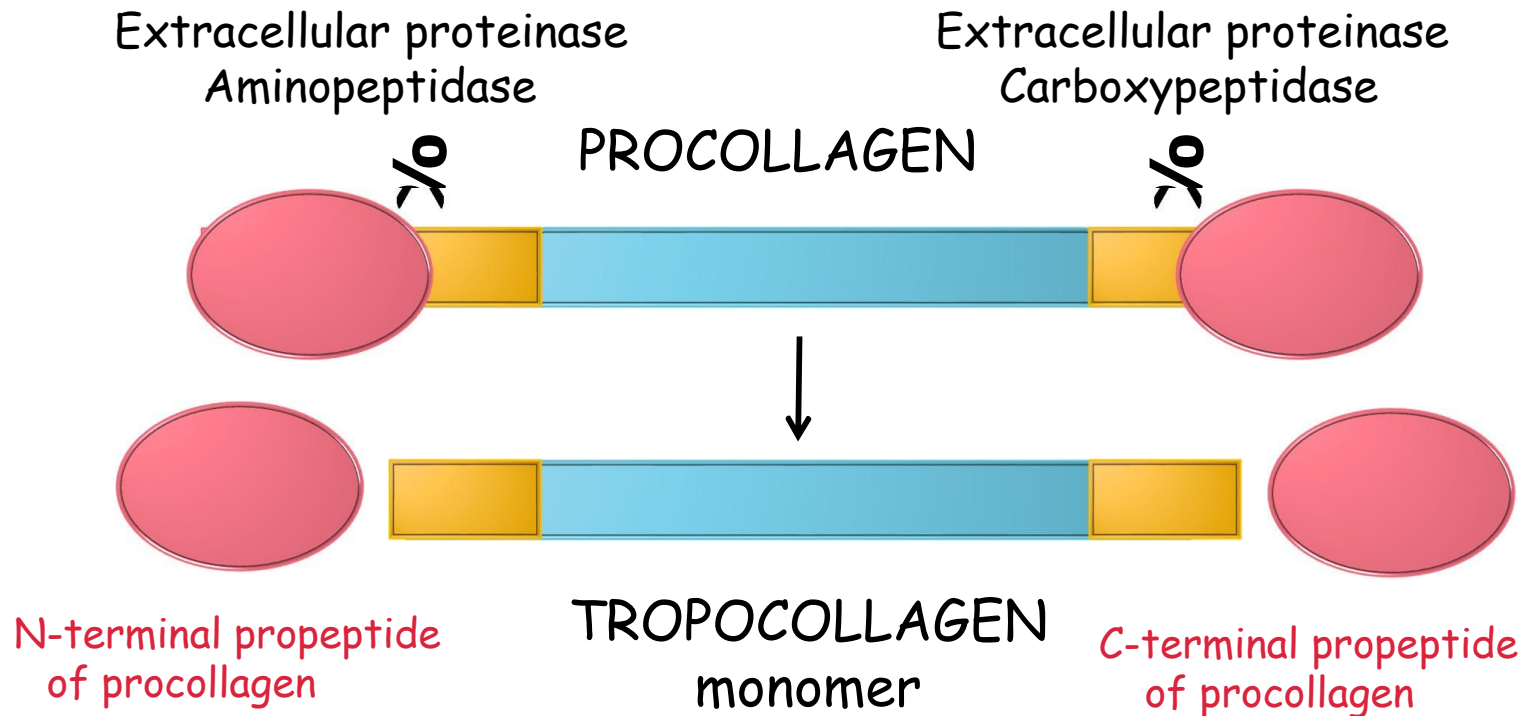
(a) Standard collagen molecule



Posttranslational modifications in the process of collagen synthesis

EXTRACELLULAR PROCESSES

CLEAVING OF THE PROPEPTIDES

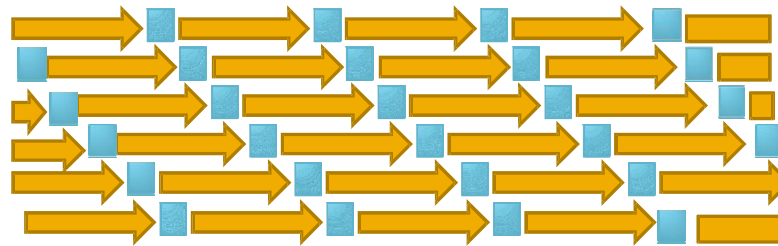


FORMATION OF FIBRILS

Tropocollagen $\longrightarrow$ Collagen fibril

The way of aggregation of fibrillary collagen

- Regular arrangement along the row and in the adjacent row
- The adjacent row is displaced by $\frac{1}{4}$ of the length



ASSEMBLING OF COLLAGEN FIBRILS

Polymerisation

Tropocollagen



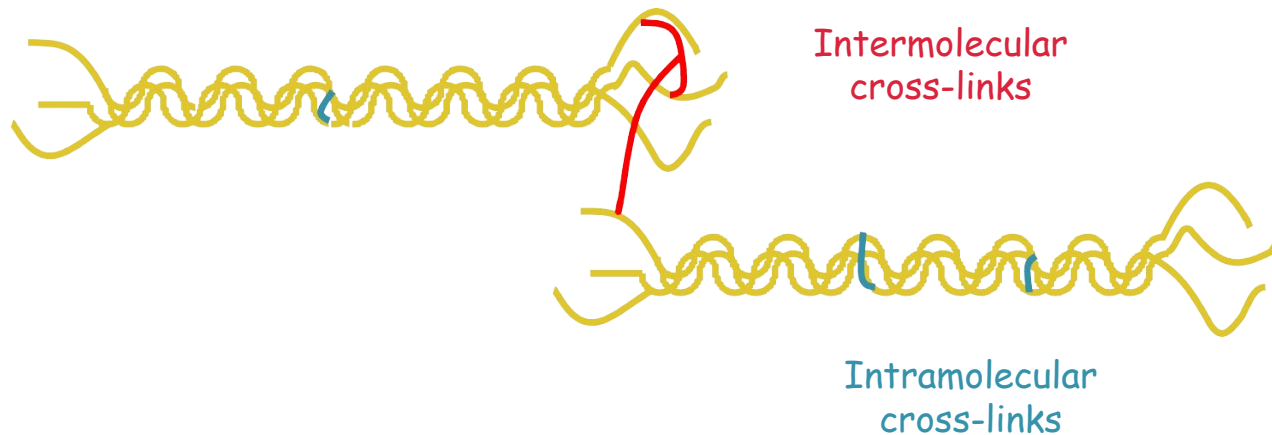
Collagen fibrils



Collagen fibers

FORMATION OF CROSS-LINKS

Collagen fibers are stabilized by formation of the covalent cross-links, which can be formed either **within** the tropocollagen molecule between the three chains – **intramolecular cross-links** and **between** the tropocollagen molecules – **intermolecular cross-links**.



FORMATION OF CROSS-LINKS

Function of cross-links

stabilization and strengthening of collagen fibril



Cross-linking

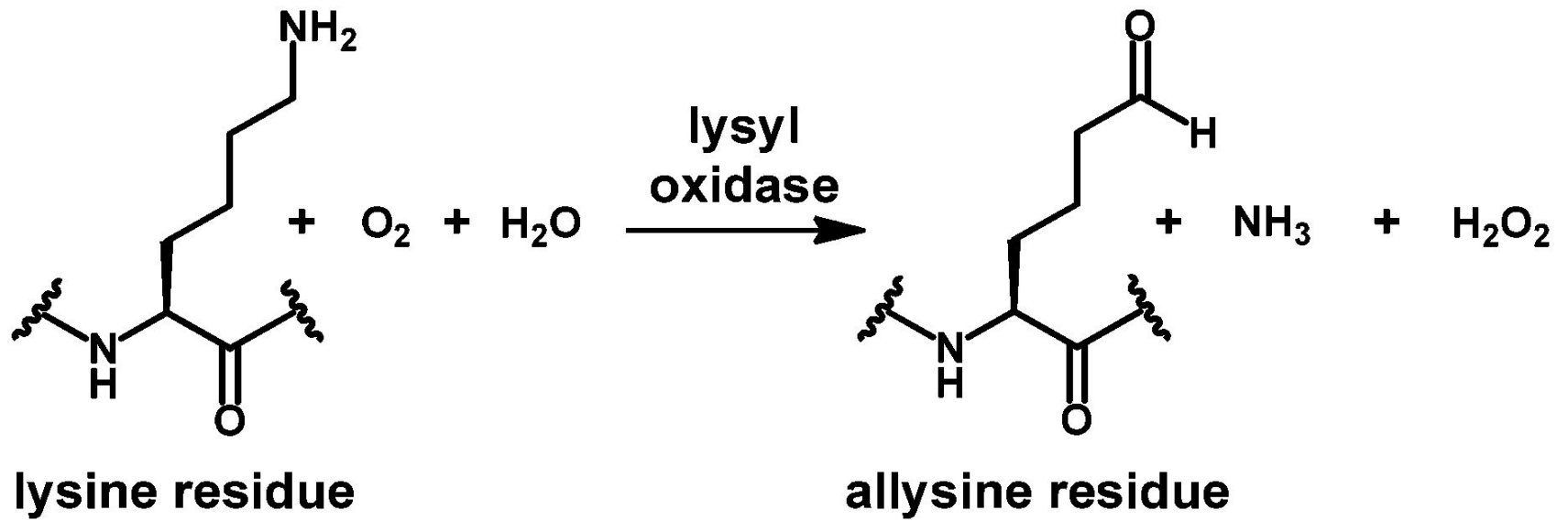
high breaking strength
lower extensibility

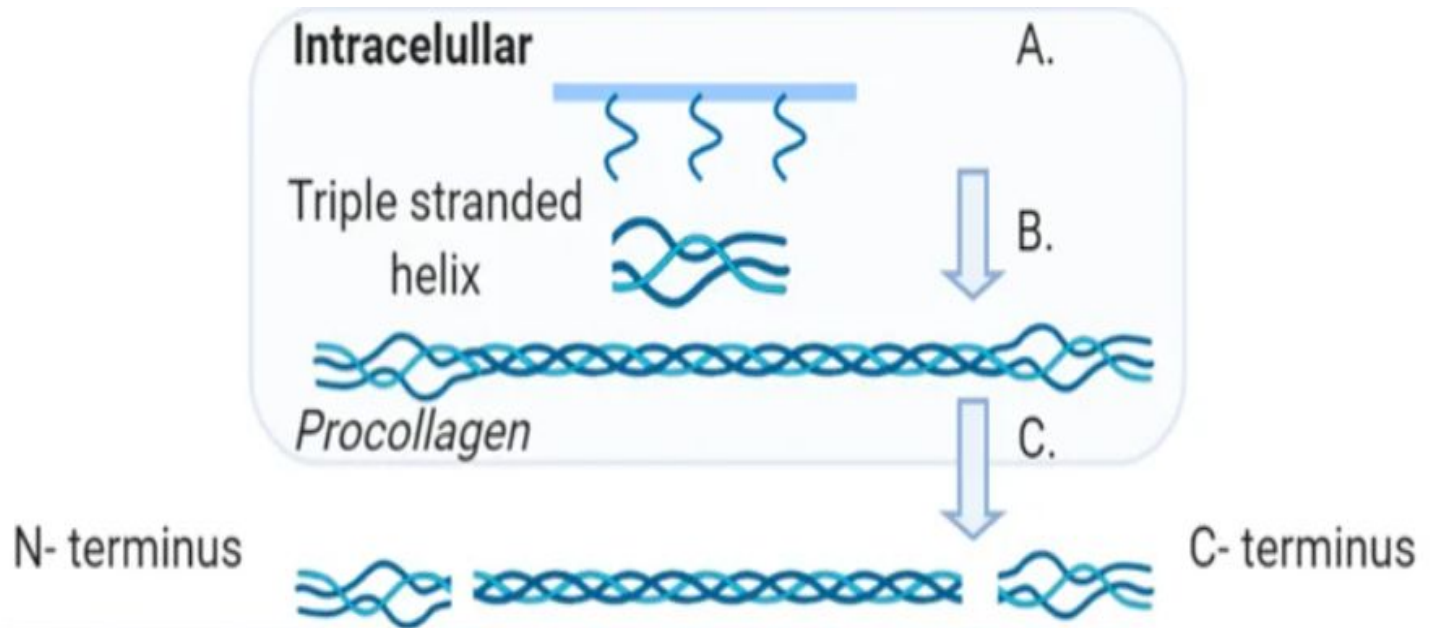
ALDOL CROSS-LINK

- Mechanism of formation

Oxidative deamination of lysine, aldehyde formation

- by the enzyme **lysyloxidase**
- aminooxidase, containing Cu^{2+}
-





Extracellular

Tropocollagen

Lysine



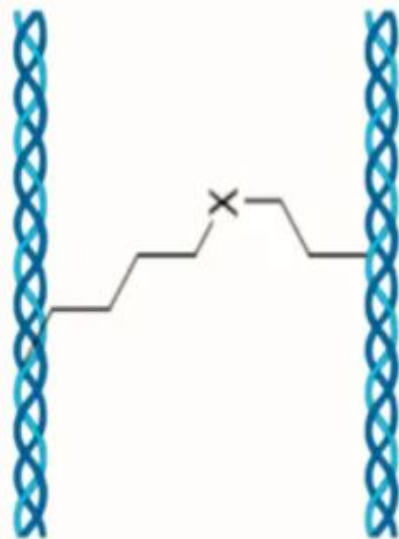
D.

LOX

Allysine



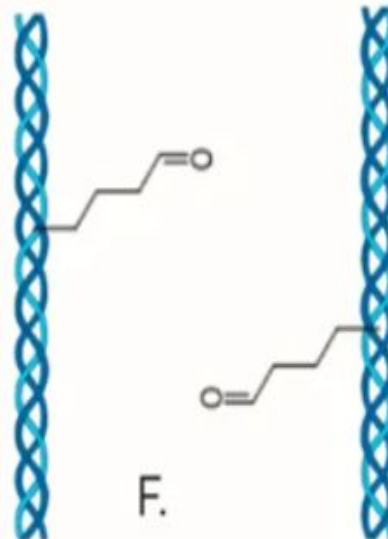
E.



*Crosslinked
Collagen*



G.

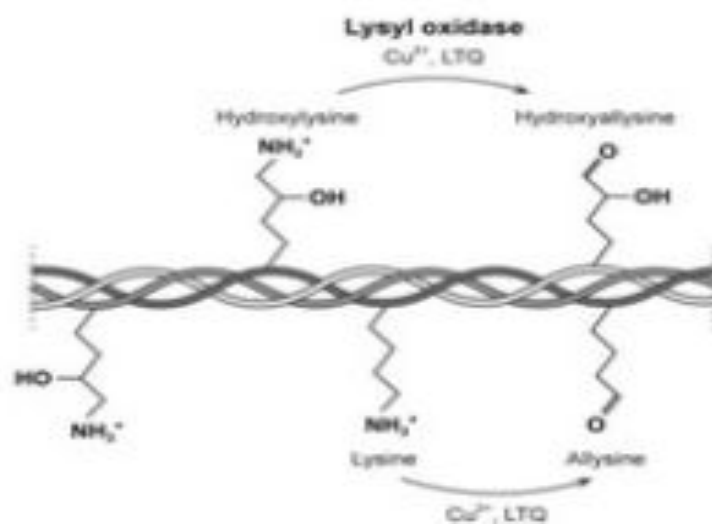


F.

*Adol
Condensation*

EXTRACELLULAR

Lysine oxidation



Collagen fibril via cross-linking



OVERVIEW OF COLLAGENS

Some fibrillar collagens

Type Molecular Occurrence structure

I	$[\alpha_1(\text{I})]_2 [\alpha_2(\text{I})]$	widely present, skin, vessels, tendons, gingiva, bone, cement, dentin, periodontal ligaments cartilage, vitreous body
II	$[\alpha_1(\text{II})]_3$	
III	$[\alpha_1(\text{III})]_3$	skin, vessels, lungs, gingiva, cement,
V	$[\alpha_1(\text{V})]_3, [\alpha_1(\text{V})]_2 \alpha_2(\text{V})$	dentin, periodontal ligaments skin, smooth muscle, bone, cement, dentin

TYPES OF COLLAGEN

Some collagens associated with fibrils

Type	Molecular structure	Occurrence
VI	$[\alpha_1(\text{VI}) \alpha_2(\text{VI}) \alpha_3(\text{VI})]$	laterally associated with collagen type II, widely present, bone, gingiva, cement, periodontal ligaments
IX	$[\alpha_1(\text{IX}) \alpha_2(\text{IX}) \alpha_3(\text{IX})]$	laterally associated with collagen type II, cartilage, vitreous body, periodontal ligaments
XII	$[\alpha_1(\text{XII})]_3$	associated with collagen type I in soft tissues, periodontal ligaments

OVERVIEW OF COLLAGENS

Some net forming collagens

Type	Molecular	Occurrence	structure
------	-----------	------------	-----------

IV	$[\alpha_1(\text{IV})]_2 [\alpha_2(\text{IV})]$
----	---

basal membranes, formation of two-dimensional net gingiva, periodontal ligaments
--

DISORDERS OF COLLAGEN SYNTHESIS

Increased collagen synthesis



- fibroses

Decreased collagen synthesis



- genetic disorder
- acquired disorders

DISORDERS OF COLLAGEN SYNTHESIS

Increased collagen synthesis - FIBROSIS

- hepatic cirrhosis
- pulmonary fibrosis
- atherosclerosis

Tissue damage stimulates collagen synthesis by fibroblasts

- e.g. damaged hepatocytes are replaced by fibrous connective tissue – hepatic cirrhosis

DISORDERS OF COLLAGEN SYNTHESIS

Decreased collagen synthesis

- Genetically conditioned
 - Ehlers-Danlos syndrome
 - osteogenesis imperfecta
- Acquired disturbances
 - Copper deficiency
 - vitamin C deficiency

ELASTIN IN ECM

Elastin is the main protein of elastic fibers, providing elasticity and recoil to the tissues.

Found in tissues requiring flexibility.

Contains desmosine cross-links.

Less abundant in dental hard tissues but important in soft tissues.

ELASTI N

Occurrence

- Arteries, particularly in aorta
- Skin
- Elastic tendons
- Lungs

ELASTIN

Properties

Extensibility and contractility

- ▣ Resembles the rubber
- ▣ After extension elastin is able to return to original size and original form
- ▣ Tensile strength is lower than in collagen
- ▣ Hydrophobic, practically insoluble in aqueous solutions



Amino acids:

- Small and nonpolar (e.g., glycine, alanine, and valine).
- Also rich in proline and lysine but contains few hydroxyproline and hydroxylysine.
- *Lysine takes part in forming of cross - links.*

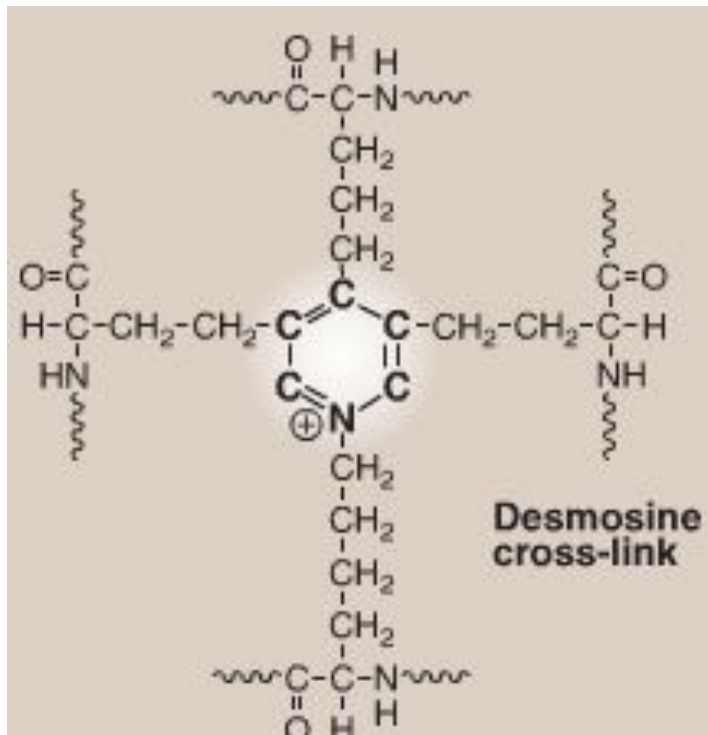
CROSS-LINKS IN ELASTIN

- Key step is an oxidative deamination of some lysine residues by copper-containing lysyloxidase (the same enzyme as in formation of cross-links in collagen).
- Cross-links may be formed within one polypeptide chain or between 2 – 4 chains

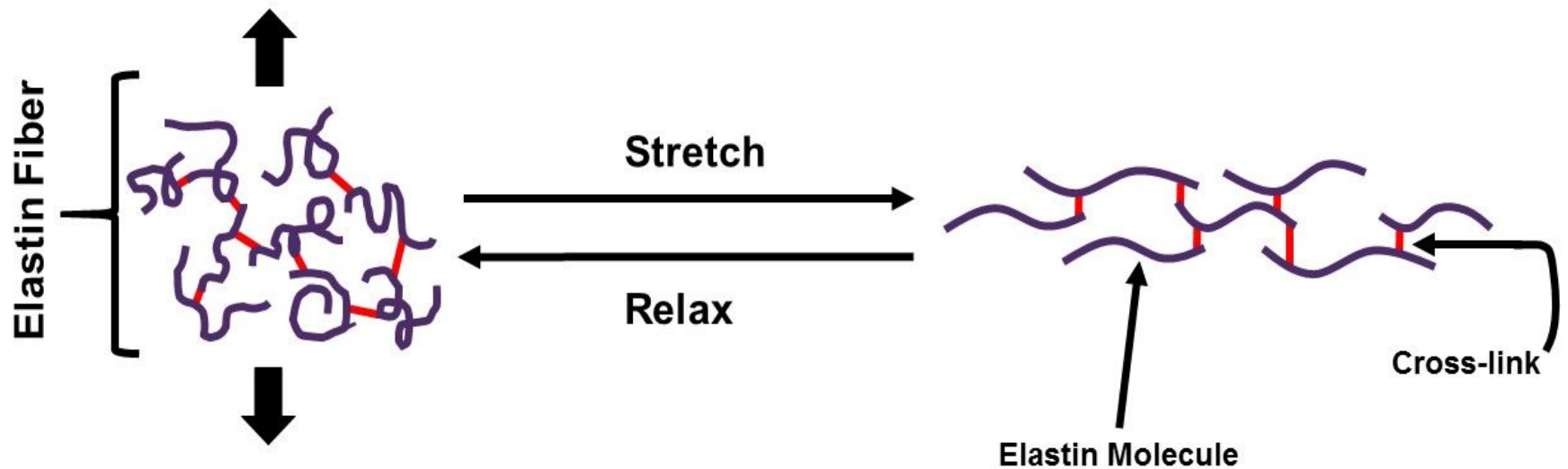
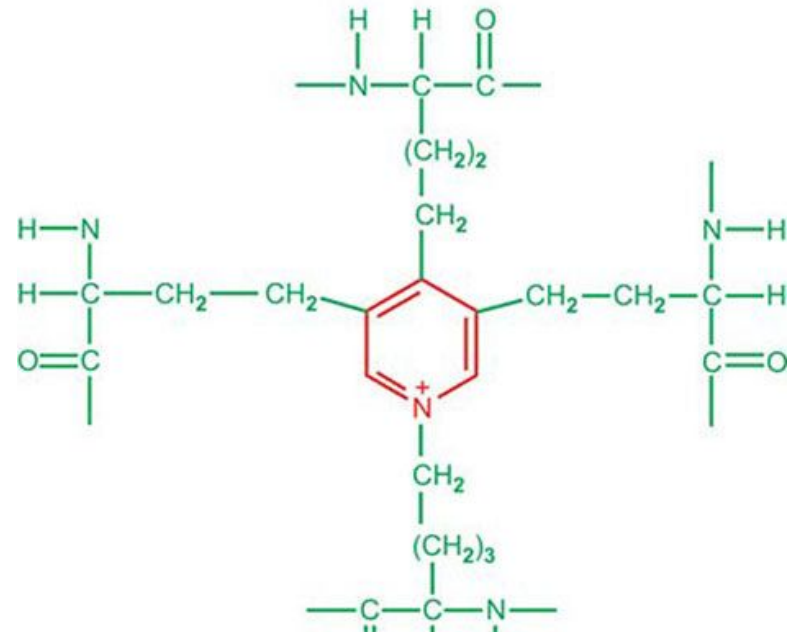
DESMOSINE

- Cross-link completely specific for elastin
- Arises from 4 side chains of LYSINE (3 oxidized and 1 nonoxidized)
- Determines the high elasticity of elastin

Linking of polypeptide chains of elastin by cross-links constitutes a three-dimensional netting explaining the “rubber-like” properties of elastin.



3 Allysine + Lysine $\xrightarrow{\text{Condensation}}$ Desmosine



ELASTIN SYNTHESIS

SYNTHESIS OF POLYPEPTIDE CHAIN

Intracellular
processes

HYDROXYLATION OF PROLINE



RESIDUES SECRETION OF



TROPOELASTIN



Tropoelastin

(globular structure, $M_r = 70\ 000$)

Extracellular
processes

Formation of cross-links



Three-dimensional netting



GROUND SUBSTANCE

Gel-like material filling spaces between fibers.

Rich in proteoglycans and GAGs.

Highly hydrated due to negative charges.

Facilitates diffusion of nutrients and metabolites.

HETEROPOLYSACCHARIDES OVERVIEW

Polysaccharides composed of two or more different monosaccharides.

Usually present as repeating disaccharide units.

Often negatively charged due to sulfate or carboxyl groups.

Major examples: glycosaminoglycans (GAGs).

GLYCOSAMINOGLYCANS (GAGS)

Long, unbranched heteropolysaccharides.

Composed of repeating disaccharide units.

Highly negatively charged → attract water.

Provide resilience and shock-absorbing properties.

TYPES OF GAGS

Hyaluronic acid (non-sulfated).

Chondroitin sulfate.

Dermatan sulfate.

Keratan sulfate.

Heparan sulfate and heparin.

PROTEOGLYC ANS

Proteoglycans are formed by
glycosaminoglycans, attached to core protein.

INTRODUCTION

Core protein + multiple GAG chains.

Form large aggregates in ECM.

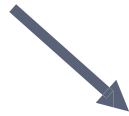
Provide compressive strength.

Important in cartilage, dentin, and periodontal tissues.

PROTEOGLYCA

NS

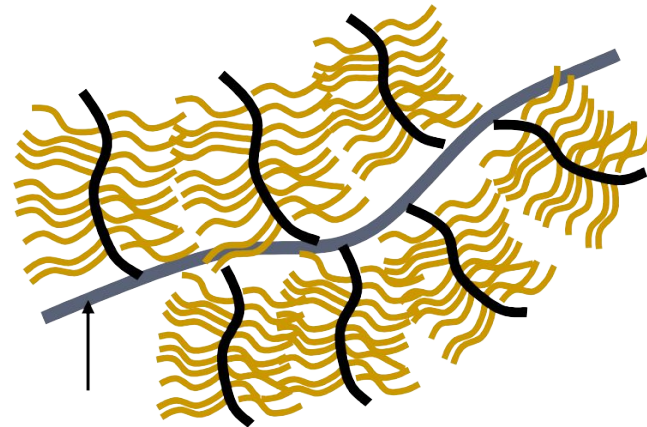
Glycosaminoglycan
(non branched saccharide
chain)



Proteoglycan
(glycosaminoglycans
and core protein)



Proteoglycan aggregate
(proteoglycans
and hyaluronate)



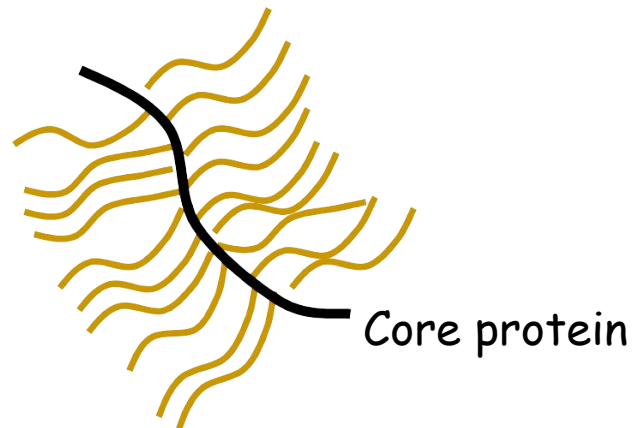
Parts of

proteoglycans

- Glycosaminoglycans (polysaccharides)
- Protein

95 %
5 %

Glycosaminoglycans



FUNCTION OF PROTEOGLYCANS

- Increase of the pressure resistance
- Sieve for macromolecules – restriction of their diffusion
- Lubrication effect
- Hydration of joint cartilages
- Adhesion of cells and their migration
- Involvement in the development of cells and tissues
- Binding of signal molecules
- In bone tissue – binding of calcium salts

PROTEOGLYCAN

- Proteoglycans are characterized by structural diversity:
 - different core proteins
 - different *GAG* chains
 - different length of *GAG* chains
- Proteoglycans differ also in localisation:
 - proteoglycans attached to basal membrane
 - interstitial proteoglycans

SELECTED PROTEOGLYCANS

Proteoglycan	Typ GAG	Function
Versican	chondroitinsulphate dermatansulphate	<i>forms proteoglycan aggregates with hyaluronate</i>
Aggrecan	chondroitinsulphate keratansulphate	<i>• cartilage, gingiva</i>
Decorin	chondroitinsulphate dermatansulphate	<i>• binds to collagen • belongs to a group of small proteoglycan rich in leucine</i>
Perlecan	heparansulphate	<i>• present in basal membrane • long core protein • forms a barrier limiting penetration of macromolecules through</i>

ADHESION GLYCOPROTEINS

Ensure specific interactions between cells and molecules of extracellular matrix.

ADHESION GLYCOPROTEINS

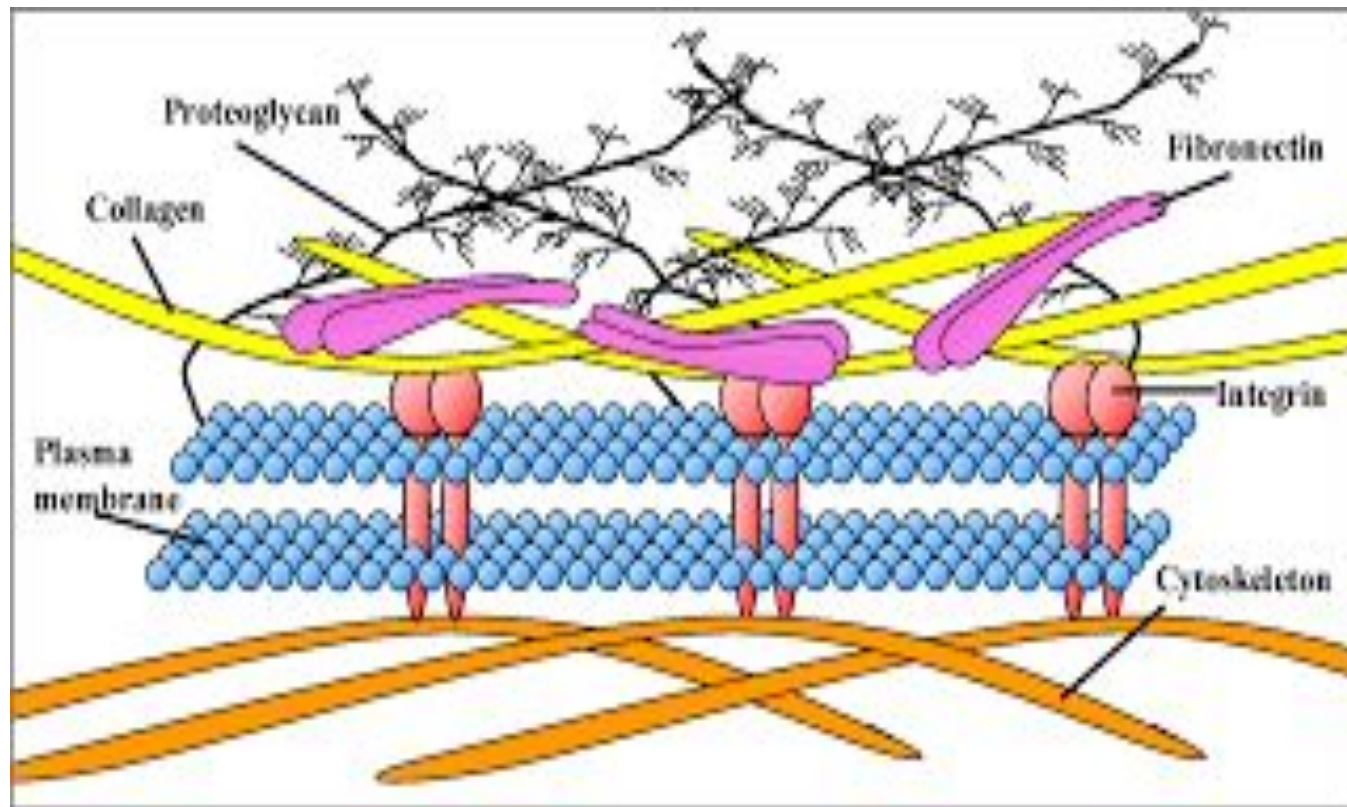
- Functions of adhesion glycoproteins
 - attachment of cells to extracellular matrix
 - organization of the compounds of extracellular matrix
- Long flexible molecules with several binding sites for:
 - collagen
 - other matrix proteins
 - polysaccharides
 - cell receptors (integrins – cell adhesion receptors)

- 
- Selected representatives of adhesion glycoproteins
 - Fibronectin
 - Laminin
 - Osteonectin
 - Chondronectin

ADHESION GLYCOPROTEINS

- Fibronectin

- Formed by two subunits arranged to the shape of letter V
- Binding sites for:
 - collagen,
 - heparansulphate,
 - hyaluronate
 - integrins
- Functions as a connection of cells in extracellular matrix containig fibrillary collagen



ADHESION GLYCOPROTEINS

- Laminin

- Binding sites for:
 - collagen of type IV
 - heparansulphate,
 - hyaluronate,
 - cell adhesion receptors
- Adhesion glycoprotein of the basal membrane - connect collagen type IV and other compounds of the membrane

SUMMARY OF ECM IN DENTAL TISSUES

Dentin: rich in type I collagen and proteoglycans.

Periodontal ligament: collagen fibers and ground substance.

Cementum and bone: mineralized ECM.

ECM remodeling is essential for tooth support and repair.



THANK YOU